

Can News Content Predict Tomorrow's Move?

origin: five US AI stocks (NVDA / MSFT / META / AMZN / GOOGL, 2021–2025)

**News content beats the next-day-direction baseline,
challenging the semi-strong EMH.**

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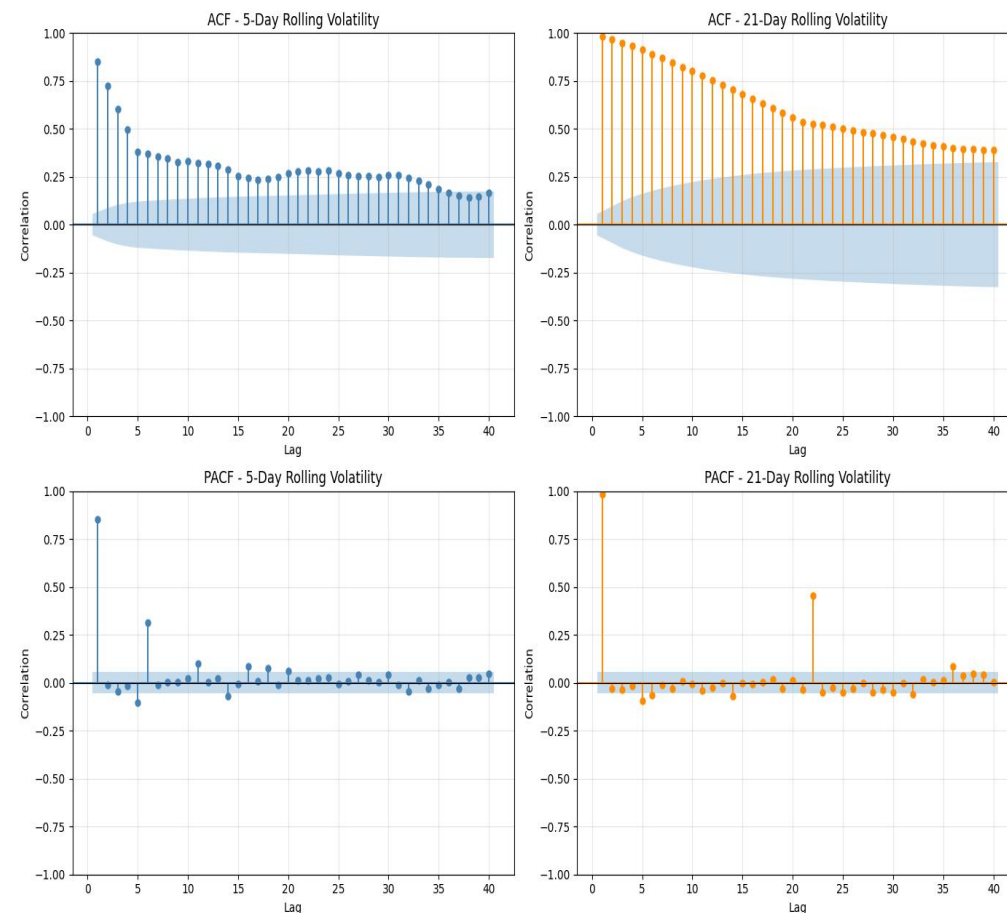
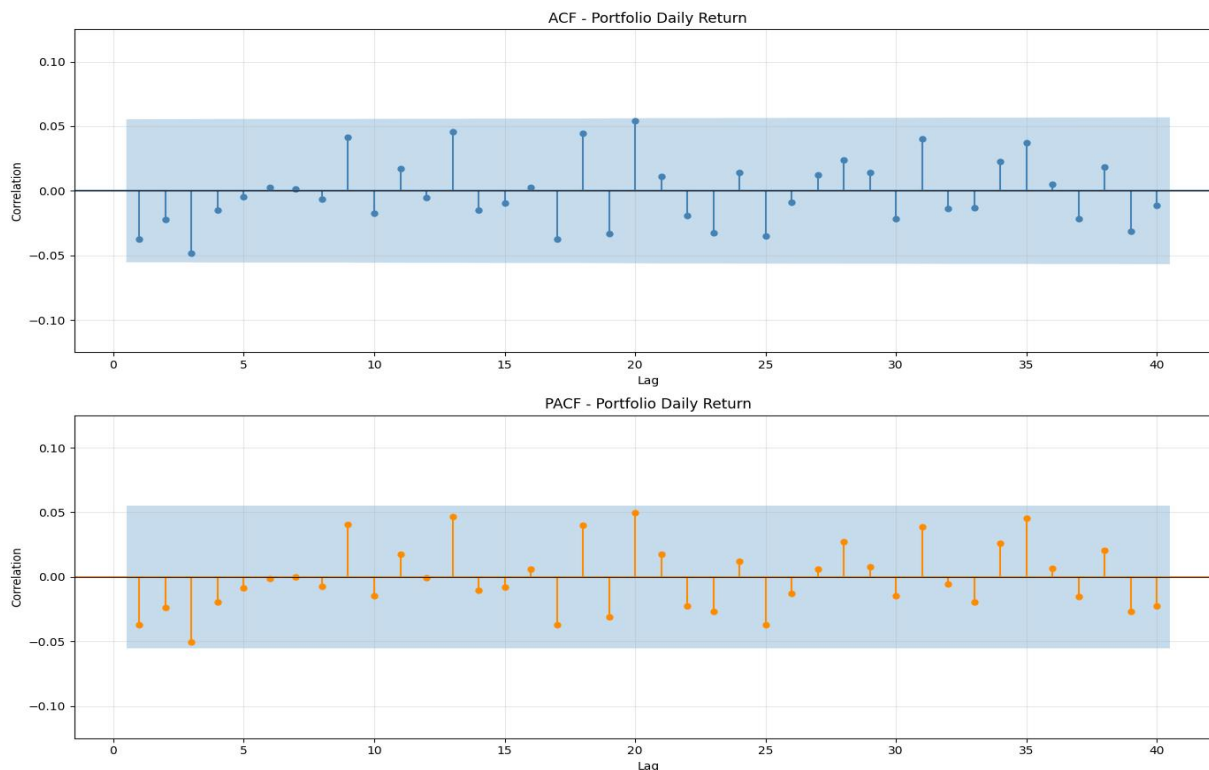




Data: Distribution & Autocorrelation

Stock	2021	2022	2023	2024	2025	Total
NVDA	60	72	335	829	1,138	2,434
MSFT	337	342	715	515	746	2,655
META	93	508	459	442	909	2,411
AMZN	1,032	810	710	608	822	3,982
GOOGL	193	207	312	315	597	1,624
Total	1,715	1,939	2,531	2,709	4,212	13,106
Share	13.1%	14.8%	19.3%	20.7%	32.1%	100%

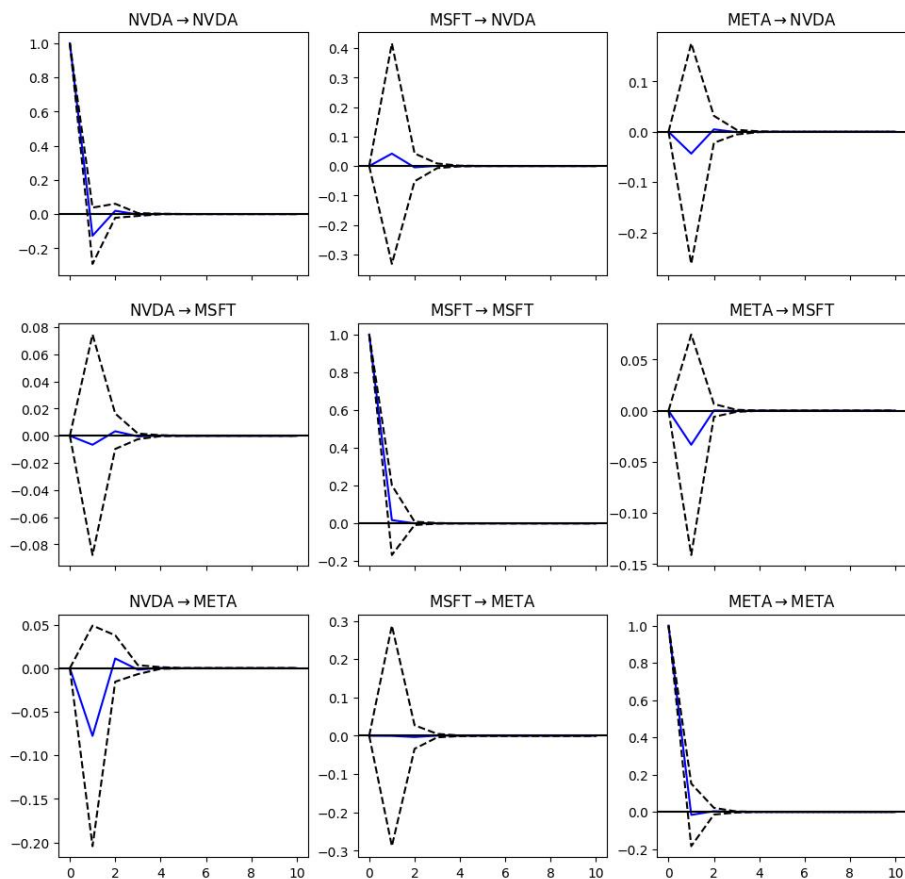
13,106 Bloomberg articles · 5 stocks · 6,270 stock-days · 2025 alone = 32.1% of all news (skewed).



Returns show almost no linear autocorrelation (unpredictable from their own past → we turn to news); volatility is highly persistent (clustering), foreshadowing the volatility U-shape.

VAR + GSADF

IRF — 3-stock (no GOOGL/AMZN) close-close, 2025 (n=249, p=1)



VAR / IRF: cross-stock linkage is weak across the full five-stock system and weakest for the AI-core three in 2025 — each stock moves mostly on its own information, a low signal-to-noise setting. (5-stock IRF omitted for space.)

GSADF: bubbles are stock-specific, not universal — only NVDA runs explosively (GSADF = 2.20, concentrated in 2024), while MSFT stays clean and the rest show only sparse episodes.



Failed Baselines

Method 1: 7-dim features (5 gated bart₋^{*} + FinBERT sign + confidence) → next-day direction; Bagging (200 depth-3 trees) & RBF-SVM; train 2021–24 / test 2025; vs majority baseline.

Result: 0 of 30 significant; Bagging train 0.60–0.67 → test 0.45–0.55 (gap +0.127); SVM's tiny gap = near-constant prediction, not signal.

Method 2 : double-0.6 gate; per-stock logit of direction on the 5 category signals; out-of-sample daily net-vote.

Per-ticker logistic regression (GLM, Binomial)

$$\Pr(\text{dir}_t = 1 \mid x_t) = \sigma\left(\beta_0 + \sum_{c \in \mathcal{C}} \beta_c x_{c,t}\right), \quad \sigma(z) = \frac{1}{1+e^{-z}}$$

$\mathcal{C} = \{\text{chip, power, algorithm, regulation, earnings}\}$; $x_{c,t} = \text{finbert_score}$ if the article passes the BART gate for category c , else 0.

Result: no ticker's CI clears its baseline; in-sample coefficients mostly insignificant and inconsistent across stocks (NVDA algorithm +0.45 vs META algorithm −0.90).

Bagging Tree & SVM

Model	Train	Test 2025	Gap	Sig.
Bagging (200×d3)	0.60–0.67	0.45–0.55	+0.127	0 / 15
RBF-SVM	—	≈ baseline	+0.002 *	0 / 15

** 0 of 30 total significant; SVM gap tiny only because it predicts near-constantly.*

GLM Logistic Regression (2025)

Ticker	Test acc	Baseline	95% CI
NVDA	0.505	0.541	[0.414, 0.595]
MSFT	0.608	0.541	[0.500, 0.716]
META	0.506	0.519	[0.392, 0.620]
AMZN	0.547	0.516	[0.422, 0.672]
GOOGL	0.500	0.568	[0.392, 0.608]

Summary

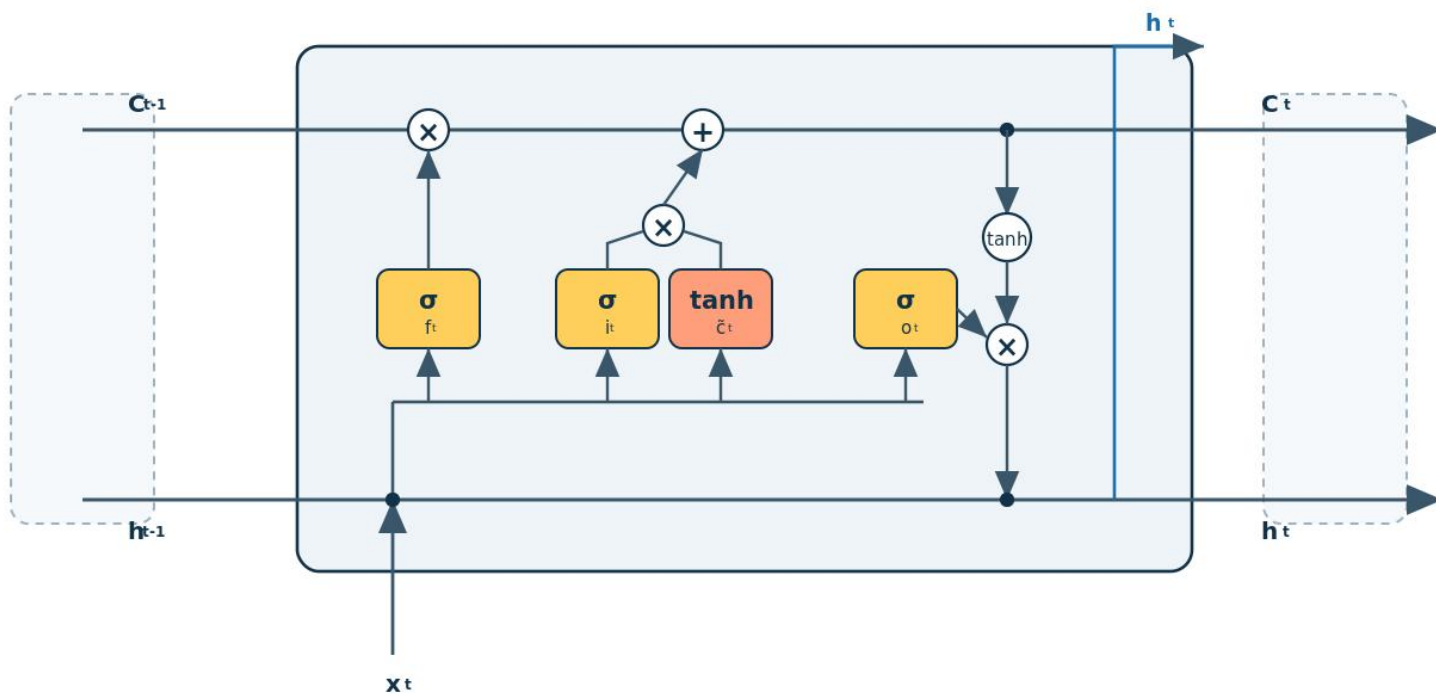
Whether flexible (tree/SVM) or linear and interpretable (logit), one global model over the full feature vector finds no out-of-sample edge — overfitting for the former, a genuinely faint signal for the latter. What's missing is conditioning on which narrative is active for which stock.

Failed Baselines

LSTM cell — propagating a $K = 10$ news-day window

Input per step: 6 features (finbert_score + 5 bart_*) + 5 one-hot ticker = 11-dim

Hidden size = 16 · window $K = 10$ · split 2021-23 train / 2024 val / 2025 test



After the last step, the final hidden state is read out:



Result (2025, out-of-sample)

test 0.522 < baseline 0.533

train 0.631 → ≈11-pt gap: overfits harder

Method 3 . We test whether the news signal is temporal: a single-layer LSTM (16 units) reads a 10-day window of the 11-dim feature and predicts next-day direction, trained 2021–23 with early stopping on 2024.

Result. It fits the training set to 0.63 but drops to 0.522 out-of-sample — below the 0.533 majority baseline, an ≈11-point train/test gap. Adding sequence memory and capacity overfits harder, not better — more model is the wrong direction in this small-sample, low-SNR regime, which points to gating rather than capacity.



Failed Baselines

One linear regression per (stock, category), then vote

$$r_{t+1} = \alpha + \beta s_{c,t} + \gamma r_t + \varepsilon_t, \quad \widehat{\text{dir}}_{t+1}^{(c)} = \mathbb{1}[\hat{\beta} s_{c,t} > 0]$$

$s_{c,t}$ = daily mean finbert_score of category c (double-0.6 gate); daily prediction = majority vote of active categories.

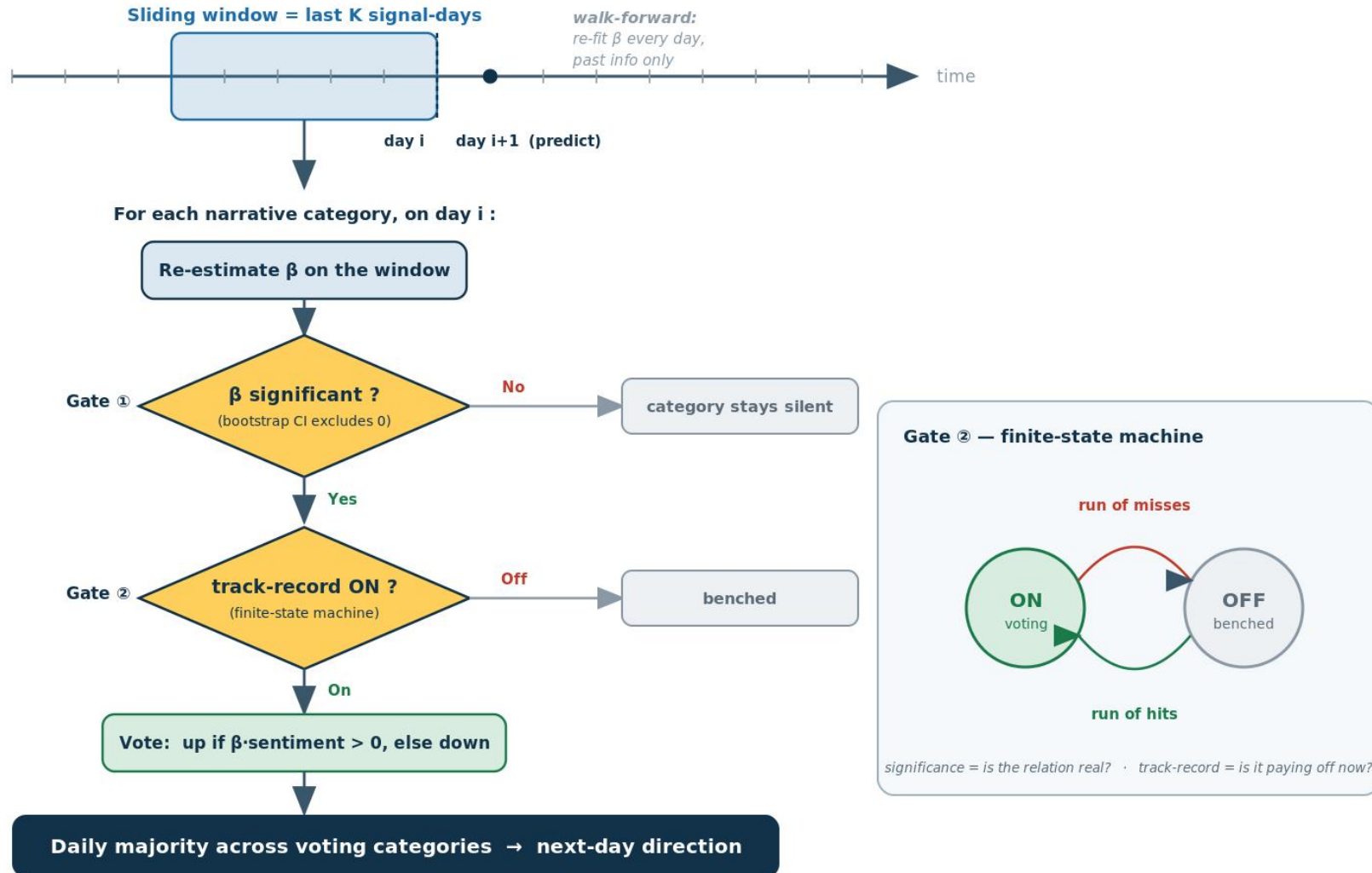
FinBERT + LP, NO gating — first excess, but unreliable (2025)

Stock	Acc	Base '24	Excess	Days	P>base
NVDA	61.5%	56.0%	+5.5%	96	88%
MSFT	58.3%	53.6%	+4.8%	60	73%
META	53.0%	53.2%	-0.1%	66	45%
AMZN	49.2%	53.2%	-4.0%	59	26%
GOOGL	45.7%	59.5%	-13.8%	70	1%

Method 4. For each (stock, category), regress next-day return on that category's daily sentiment; the sign of β says whether the narrative predicts up or down. Each day, active categories vote (up if $\beta \cdot \text{sentiment} > 0$) and the majority is the call — with β fixed from 2021–24, no gating.

Summary. A narrative-specific linear signal does exist — NVDA +5.5%, MSFT +4.8%, the first method to beat baseline. But it's unreliable: META flat, AMZN/GOOGL negative, and NVDA/MSFT fall short of a convincing bar — one fixed β applied to every day dilutes good signal with stale ones.

NG-LP — sliding window + double gate



NG-LP = Narrative-Gated Local Projections

A Local Projection is one simple linear regression that predicts the next-day return from a single narrative's sentiment. Narrative-gated means each narrative is allowed to vote only when it is currently both statistically real and actually working. So NG-LP is a set of per-narrative next-day predictors, filtered day-by-day by two gates, then combined by majority vote.

NG-LP by stock — 2025 (out-of-sample)

Universe	Acc	Base	Excess	n	P>base
NVDA	63.3%	53.6%	+9.7%	49	91%
MSFT	66.7%	52.2%	+14.5%	27	92%
META	64.5%	51.8%	+12.8%	31	91%
AMZN	66.7%	50.9%	+15.7%	3 *	75%
GOOGL	40.0%	53.7%	-13.7%	25	9%
All 5 pooled	60.0%	52.9%	+7.1%	135	96%
CORE-3 pooled	64.5%	53.5%	+11.0%	107	99%

* AMZN has only 3 voting days — too few to trust. Weak-AI GOOGL is the control that fails (-13.7%).

Volatility stratification — CORE-3 pooled (U-shape)

Volatility band	n	Acc	95% CI	P>base	mean ret
Small	36	72.2%	[58.3%, 86.1%]	99%	0.26%
Mid	35	48.6%	[31.4%, 65.7%]	31%	1.05%
Large	36	72.2%	[55.6%, 86.1%]	99%	3.68%

Accurate on both quiet & shock days (72.2%), near-random in the middle (48.6%). Bands defined ex-post.

Results. On the AI-core three (NVDA / MSFT / META), NG-LP reaches 64.5% vs a 53.5% baseline — +11.0 pts out-of-sample in 2025, with $P(\text{acc} > \text{base}) = 99\%$; each core stock beats its own baseline at $\geq 90\%$.

Control holds. The weak-AI stock GOOGL fails (−13.7%), and dropping the weak-AI pair lifts the pooled edge from +7.1% (all 5) to +11.0% (CORE-3) — the signal is specific to the AI narrative, not a blanket market effect. (AMZN's +15.7% rests on only 3 days — ignored.)

Volatility U-shape. Accuracy is highest on both quiet and shock days (72.2%) but collapses to near-random in the middle (48.6%) — the model helps when the market is in a clear directional state.



Conclusion, Limitations & Outlook

Conclusion

Against a strict 2025 full-year baseline, NG-LP beats each AI-core stock and reaches +11.0 pts pooled (64.5% vs 53.5%, $P = 99\%$) out-of-sample. This shows that news narrative structure carries next-day directional information not yet fully impounded in prices, in tension with a fully semi-strong-efficient market. We claim directional predictability only — not tradeable abnormal return (execution cost and capacity are out of scope).

In small-sample, low-SNR settings, a simple linear method beats flexible ML. With only a few hundred usable days per stock, bagging trees, SVM and the LSTM all overfit and collapse to baseline out-of-sample, while a simple linear Local Projection is what actually generalizes — model flexibility is a liability here, and the signal must come from structure (gating), not more capacity.

Limitations

News is heavily skewed — 2025 alone is ~32% of all articles, so effective information per stock-day is uneven. The 2021–2024 estimation window is short and non-stationary, spanning bubble regimes (notably NVDA 2024), which complicates stability. 2026 data are still too incomplete to extend the test forward. All hyperparameters (0.6 gates, window K , FSM thresholds) are fixed a priori or on 2024 validation — never tuned on the 2025 test set.

Outlook

Add execution cost, capacity, and tradeability tests to see whether the edge survives as a real strategy; run a forward re-test as 2026+ data arrive; and explore macro / monetary-policy regime as an extra gating condition.

— Thank you —

